

KUSTENDIL, BulgariaWMO: **156010**Lat: **42.267N**Lon: **22.767E**Elev: **527**StdP: **95.15**Time Zone: **2.00 (EUE)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.4	-9.5	-15.1	1.1	-11.4	-12.2	1.4	-8.0	10.3	1.8	8.7	1.2	0.4	N/A	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	33.6	19.9	31.7	19.5	30.1	19.1	21.3	30.5	20.5	29.5	19.7	28.3	1.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.8	25.9	17.2	13.1	24.5	16.6	12.6	23.6	64.3	30.7	61.3	29.6	58.6	28.4	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	6.7	5.1	-15.3	36.4	3.4	2.1	-17.8	37.9	-19.8	39.2	-21.7	40.4	-24.2	42.0		
			-15.6	23.1	3.2	1.0	-17.9	23.8	-19.8	24.4	-21.6	25.0	-23.9	25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	0.0	2.0	6.6	11.1	15.9	19.4	22.0	22.0	17.3	12.1	5.8	0.8
	DBStd	8.69	4.57	4.61	4.18	3.60	3.20	3.17	2.76	2.72	3.22	3.94	4.27	5.10
	HDD10.0	1132	309	224	118	28	1	0	0	0	1	25	138	287
	HDD18.3	2893	568	456	363	218	87	24	4	3	56	195	377	543
	CDD10.0	1612	1	1	14	61	184	283	372	372	221	90	12	2
	CDD18.3	331	0	0	0	1	12	57	117	117	26	2	0	0
	CDH23.3	3816	0	0	1	18	192	663	1293	1283	328	37	1	0
	CDH26.7	1441	0	0	0	3	34	218	531	551	101	4	0	0
Wind	WSAvg	1.1	0.9	1.2	1.7	1.4	1.1	1.0	1.2	1.0	0.9	0.7	0.8	0.9
Precipitation	PrecAvg	588	36	40	41	52	61	64	48	45	46	53	45	59
	PrecMax	747	90	101	84	117	120	123	103	109	119	122	97	145
	PrecMin	329	4	3	6	14	36	11	7	8	16	5	0	15
	PrecStd	119	24	26	21	24	23	30	26	30	26	34	23	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	17.1	21.5	25.9	29.7	33.2	36.3	35.9	32.1	26.7	20.1	16.0
		MCWB	8.6	9.8	12.3	15.2	18.1	20.5	19.9	20.3	18.5	17.0	13.7	11.8
	2%	DB	10.5	14.0	18.8	22.3	27.3	31.1	33.6	33.8	29.5	24.0	17.1	13.0
		MCWB	6.9	8.2	10.6	12.9	17.1	20.1	19.9	20.0	17.8	15.5	12.2	9.3
	5%	DB	8.1	11.4	16.6	20.1	25.5	29.2	31.6	31.8	27.2	21.5	14.5	10.4
		MCWB	5.3	7.0	9.6	12.1	16.6	19.5	19.7	19.3	17.0	14.5	10.8	8.1
	10%	DB	6.2	9.1	14.3	18.3	23.6	27.2	29.7	29.9	24.7	19.3	12.5	7.9
		MCWB	4.1	5.7	8.5	11.3	15.9	18.7	19.2	18.8	15.9	13.5	9.4	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	10.1	12.9	15.4	19.4	22.0	22.5	22.5	20.0	17.7	14.2	12.4
		MCDB	13.1	16.2	21.0	24.4	27.3	30.4	32.3	32.4	28.7	25.5	19.2	15.6
	2%	WB	7.2	8.5	11.2	13.7	17.9	20.9	21.2	21.2	18.4	16.2	12.6	9.7
		MCDB	10.0	13.2	17.8	20.7	25.3	29.6	30.4	31.0	27.3	22.2	16.2	12.3
	5%	WB	5.7	7.2	9.9	12.7	17.0	20.0	20.4	20.2	17.5	15.1	11.3	8.1
		MCDB	7.7	11.1	15.8	18.9	24.4	28.1	29.2	29.7	25.3	20.4	14.1	10.5
	10%	WB	4.2	5.9	8.7	11.7	16.1	19.0	19.6	19.3	16.7	14.0	9.7	6.2
		MCDB	6.1	8.9	13.7	17.4	23.0	26.5	28.1	28.2	23.2	18.5	12.3	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.7	9.8	11.1	11.3	12.2	12.9	13.8	14.1	12.9	11.9	9.4	6.7
		MCDBR	10.4	14.8	15.9	15.7	16.3	16.9	17.6	17.9	17.5	15.8	12.3	9.0
		MCWBR	7.1	9.5	8.9	7.9	7.6	7.2	6.4	6.6	7.3	8.2	7.8	6.2
	5% WB	MCDBR	9.0	13.7	14.8	13.7	14.8	15.4	15.3	16.0	15.1	13.7	11.0	8.5
		MCWBR	6.3	9.0	8.5	7.3	7.4	7.1	6.5	6.6	6.8	7.6	7.1	6.0
Clear-Sky Solar Irradiance	taub	0.313	0.341	0.384	0.441	0.434	0.424	0.424	0.437	0.412	0.374	0.341	0.315	
	taud	2.387	2.318	2.238	2.091	2.183	2.249	2.250	2.213	2.264	2.359	2.413	2.410	
	Ebn at Noon	825	854	854	827	844	852	847	823	816	806	783	788	
	Edh at Noon	80	101	124	154	144	136	134	135	119	96	79	73	
All-Sky Solar Radiation	RadAvg	1.64	2.51	3.53	4.48	5.40	6.19	6.43	5.76	4.24	2.99	1.93	1.41	
	RadStd	0.25	0.33	0.40	0.53	0.42	0.52	0.45	0.46	0.35	0.36	0.19	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)	+0.48	N/A	N/A	N/A	+0.36	N/A	N/A	-140	N/A	N/A

Nomenclature: See separate page